Testing the Biomarker-by-Drug Interaction When the Carryover Half-Life Is Chosen From the Data

2026-10-09 09:54 PDT

Author. pmsimstats team

Purpose. The recommended analysis of the biomarker-by-drug interaction in an aggregated N-of-1 (Hybrid) trial, F1 (`docs/56`), weights each off-drug visit by an exposure-decayed drug indicator $D_{bc,it}$ whose half-life $t^a_{1/2}$ is set by the analyst. With $t^a_{1/2}$ fixed in advance the test holds its size, even when the value is wrong. Face validity, however, asks that the analysis not use a half-life the trial's own data contradict, and the project decided on 2026-10-09 that the half-life should be chosen from the data. This note sets out how to do that without losing control of the Type I error. It explains why the one data-driven variant already run (F1A, AIC over a grid with the interaction in the model) is liberal, derives the mechanism, and presents eight procedures and a permutation approach, each with its idea, an exact procedure, a validity argument and its conditions, its cost, what it reports and its status. Two worked examples, one trial and a small Monte Carlo illustration on stored stress-test trials, show what each procedure does. A decision table and a provisional plan close the note. It is written for trial statisticians and methodologists, including the authors of the original `pmsimstats` work [1], and assumes no other pmsimstats-ng document beyond the model statement of `docs/56`.

Contents

Notation index and glossary 4
1. Summary for the busy reader 5
2. Scope and relation to `docs/56` 6
3. The problem 6
3.1 Where the half-life enters 6
3.2 The analysis half-life is a working parameter 7
3.3 Why a fixed half-life keeps its size even when it is wrong 7
3.4 Why choosing the half-life by AIC inflates size 7
3.5 How much the data say about the half-life 8
4. Methods 9
4.1 Method 1: fixed in advance, with a sensitivity grid 9
4.2 Method 2: selection blind to the interaction (F1A0) 10
4.3 Method 3: sample splitting and cross-fitting 11
4.4 Method 4: reporting the half-life 12
4.5 Method 5: parametric bootstrap of the whole procedure (F1AB) 13
4.6 Method 6: joint estimation of the half-life as a continuous parameter 14
4.7 Method 7: the maximum statistic over the grid 14
4.8 Method 8: marginalizing over the half-life 15
4.9 Permutation tests 16
5. Worked example 1: one null trial 16
5.1 Set-up 16
5.2 The profile and the test across the half-life 16
5.3 What each method reports 17
6. Worked example 2: a small Monte Carlo illustration 18
6.1 Set-up 18
6.2 Results 19
6.3 Reading 21
6.4 Run time 22
7. Recommendation 22
7.1 Decision table 22
7.2 Provisional plan 23
7.3 The stress-test extension needed to confirm it 24
8. Limitations 24
9. How to reproduce 24
10. Reproducibility 25
11. References 25

List of Tables

1 Notation index for this note 4

2 Null rejection rates of F1 (fixed 0.5 week) and F1A on the same trials of the 11 non-prognostic null cells, by the half-life F1A selected (verified from `replicates-full.rds` and `replicates-full-extra2.rds`; stored setting) 8

3 Share of null trials in which F1A's AIC chose each analysis half-life (weeks), selected cells, 1,000 trials each, and pooled over all cells and hypotheses (verified; stored setting) 8

4 Example 1: ML log-likelihoods with and without the interaction, the likelihood-ratio statistic, and the REML + CR2 test of the interaction at each grid half-life (verified) 16

5 Example 1: what each procedure reports for the same null trial (verified) 17

6 Example 2: rejection rates at nominal 0.05 (binomial MCSE in parentheses) on stored stress-test trials, stored setting (verified; the warp-speed MCSEs understate the error, and their pooled column is the mean of the two cell rates) 19

7 Example 2: warp-speed bootstrap critical values for $|T|$, the 95% quantile of the pooled resampled statistics of each cell (verified) 20

8 Example 2: share of trials choosing each analysis half-life (weeks); F1A and F1A0 agreed in 86.6% and 89.4% of the trials of these two cells (verified) 21

9 Decision table for choosing the analysis half-life from the data (size and power from the illustration of Section 6, stored setting) 22

List of Figures

1 Example 1, one null Hybrid trial ($N = 70$, true $t_{1/2} = 0.5$ week, refined process). A: ML deviance $2\{\ell_{\max} - \ell(h)\}$ of F1 with the interaction (solid) and without it (dashed) over a fine grid of analysis half-lives; the horizontal line is $\chi^2_{1,0.95} = 3.84$, and points mark the grid $\mathcal{T}_a$. B: the CR2 t-statistic of the interaction across the same half-lives, with ± 1.96 marked. 17

2 Example 2: rejection rates at nominal 0.05 with 2-MCSE bars, by procedure, in the four stress-test cells (stored setting). Circles: nominal reference distribution; triangles: warp-speed bootstrap calibration. The vertical line marks 0.05 in the null cells; the alternative cell (top right) shows power. 20

Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`). Symbols marked *local* are defined here for this note only. (Notation, 2026-10-09: the centered, unscaled biomarker of the code column `bmc`, formerly written b_i and, in Section 4.2, $\tilde{b}_i$, is now the canonical B_i^c; b_i is kept for the standardized biomarker. The time term is $\beta_T w_t$, the normal law $\mathcal{N}$, and the compound-symmetry correlation ρ_{cs}; probability is Pr (P is the number of paths). Symbols previously marked local or proposed that `NOTATION.md` now lists are marked canonical.)

Table 1: Notation index for this note

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t; $t = 0$ is the baseline row (code `Sx`)	canonical
B_i, B_i^c	biomarker; biomarker centered at the trial's baseline mean, $B_i - \bar{B}$, not scaled (code `bmc`). Test statistics are invariant to the scaling, so the results hold equally for the standardized b_i	canonical
D_{it}	binary drug state: 1 on drug, 0 off	canonical
t_{sd}	time since discontinuation, cumulative within an off-drug run, 0 on drug and before first exposure (code `tsd`)	canonical
$t_{1/2}$	true carryover half-life of the data-generating process, in weeks	canonical
$t_{1/2}^a$	half-life assumed by the analysis in $D_{bc,it}$, in weeks	canonical
h	a candidate value of $t_{1/2}^a$, used as a function argument	local
$D_{bc,it}(h)$	exposure-decayed drug indicator at analysis half-life h: 1 on drug, $2^{-t_{sd}/h}$ off drug after first exposure, 0 before it; $h = 0$ gives D_{it} (code `Dbc`, `X` in the driver)	canonical
η_t	expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
$\beta_{bm:D}$	the biomarker-by-drug interaction, the estimand	canonical
β_D	coefficient of $D_{bc,it}$, the drug main effect	canonical
β_0, β_B, β_T	intercept, biomarker main effect and linear-time coefficient	canonical
β_η, $\beta_{bm:\eta}$	coefficients of η_t and $B_i^c \eta_t$	canonical
u_i, σ_u^2	participant random intercept and its variance	canonical
ε_{it}, σ_ε	observation-level residual and its standard deviation	canonical
ρ_a	working CAR(1) correlation per week of the analysis model	canonical
$\mathcal{T}_a$	the five-point grid $\{0, 0.25, 0.5, 1, 2\}$ weeks (compendium, eq:invd-aic)	local
$\ell_1(h)$, $\ell_0(h)$	maximized ML log-likelihood of F1 at h, with and without the interaction term	local
$\hat{h}_1$, $\hat{h}_0$	$\arg\max_h \ell_1(h)$ (the F1A choice) and $\arg\max_h \ell_0(h)$ (the F1A0 choice)	local
$T(h)$	CR2 t-statistic for $\beta_{bm:D}$ in F1 refitted by REML at h	local
$\mathrm{LR}(h)$	$2\{\ell_1(h) - \ell_0(h)\}$, the likelihood-ratio statistic for the interaction at h	local
$\mathrm{AIC}_1(h)$, $\mathrm{AIC}_0(h)$	AIC of F1 at h with and without the interaction term	local
$a_i(h, Y)$	participant i's contribution to the exposure contrast after partialling (Section 4.2)	local
$g(Y)$	a half-life selection rule that uses the outcome and the design only	local
h_A, h_B	half-lives chosen on the two halves of a split sample	local
ω	$\log t_{1/2}^a$, the half-life as a continuous parameter (Section 4.6)	local
$D_{bc,i}$, V_i	participant i's vector of $D_{bc,it}$ and working covariance	local
$I_{\beta\omega}$, U_β	blocks of the information matrix and of the score	local
q, c_q	randomization-path index (canonical), and a path-level constant (Section 4.6, local)	canonical, local
W_i, $\hat{\xi}$	participant i's null-model design matrix and the fitted null-model fixed effects (Section 4.5)	local

Symbol	Meaning	Status
c_{bm}	biomarker-BR coupling in the data-generating process (the predictive effect)	canonical
ρ	AR(1) serial correlation per week	canonical
N	total sample size across paths	canonical
M, m	number of bootstrap resamples per trial, and the resample index	local
T_m^*, p^*	bootstrap statistic and bootstrap p-value	local
n_{sim}	number of Monte Carlo replicates in a cell	canonical
$c_{bm,TV}$	biomarker-natural-course coupling of a prognostic cell	canonical
α	nominal level, 0.05	canonical

Glossary.

- **Hybrid design.** Eight visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 plus baseline, open-label titration, blinded discontinuation and a brief crossover, four randomization paths [1].
- **F1.** The recommended analysis (`docs/56`): linear time, $D_{bc,it}$ at $t^a_{1/2}$, the expectancy terms η_t and $B^c_i\eta_t$, a random intercept, continuous-time AR(1) residuals (`corCAR1`) and CR2 cluster-robust inference with Satterthwaite degrees of freedom [3, 4].
- **F1A.** F1 with $t^a_{1/2}$ chosen by AIC over $\mathcal{T}_a$, ML for the choice, REML refit, then tested as if fixed (the stress-test variant of `docs/47`).
- **F1A0.** F1 with $t^a_{1/2}$ chosen by AIC over $\mathcal{T}_a$ on F1 *without* the interaction term, then refitted with it (proposed in `docs/56`; illustrated here).
- **F1AB.** F1A, or F1A0, calibrated by a parametric bootstrap from the fitted null model that repeats the selection in every resample.
- **Size.** The null rejection rate at nominal α; the stress test caps any cell at 0.075 and asks the pooled rate to lie in the 2-MCSE band around 0.05 (`docs/47`).
- **Base cell.** The stress-test cell with $N = 70$, AR(1) at $\rho = 0.7$, $t_{1/2} = 0.5$ week, no dropout and no prognostic pathway (`docs/46`, `docs/47`).
- **Working parameter.** A parameter of the analysis model that need not correspond to any parameter of the data-generating process; here $t^a_{1/2}$ (Section 3.2).

Evidence labels. Each claim carries one of five labels: *quoted* (text reproduced from a source), *inspected* (read from source code or stored output), *derived* (follows algebraically, possibly to first order, as stated), *verified* (computed by running code, with the script named in Section 10) or *argued* (a reasoned claim that has not been derived in full or checked by computation).

1. Summary for the busy reader

The question. F1 tests $\beta_{bm:D}$ with $D_{bc,it}$ built at a half-life $t^a_{1/2}$. Fixed in advance at 0.5 week, F1 rejected 0.051 of 11,000 null trials pooled over the 11 non-prognostic null cells, and at most 0.062 in any cell, including the cells whose true half-life was 0 or 1 week (quoted from `docs/47`, Section 3; stored setting: configuration C at $\rho = 0.7$, the published recursive carryover and fixed allocation; an anchored rerun is pending). The project now wants the half-life chosen from the data.

Why the obvious way fails. F1A chose $t^a_{1/2}$ by AIC with the interaction in the model and rejected 0.057 pooled (outside the band 0.046 to 0.054) and 0.073 at true $t_{1/2} = 1$ week (quoted, `docs/47`). Because the candidates have equal parameter counts, $\text{AIC}_1(h) = \text{AIC}_0(h) - \text{LR}(h)$: the rule rewards the half-life at which the interaction looks strongest, and the test then ignores that it was chosen (derived, Section 3.4). On the stored trials, when F1A chose 2 weeks it rejected 0.079 against F1's 0.046 on the same trials (verified).

What the data say about the half-life. Coarsely something. AIC chose 0 in 87% of null trials when the truth was 0, and 1 or 2 weeks in 77% when it was 1 week, but the true 0.5 week in only 38% of base-cell trials, and 1 or 2 weeks in 85% of compound-symmetry trials whose truth was also 0.5 (verified from

stored replicates). In the example trial the profile likelihood cannot separate 0 from 0.5 week but rejects a week or more decisively (verified, Section 5). The analysis half-life is also a working parameter, not the pharmacological one (Section 3.2).

What to do instead. Choose the half-life by a criterion that does not see the interaction (F1A0), or on data the test does not use (sample splitting and cross-fitting), or calibrate the whole select-then-test procedure (F1AB); report the half-life with a profile-likelihood interval. Under the null with the biomarker independent of the outcome, a selection that uses only the outcome is harmless (derived to first order, Section 4.2), and the half-life is asymptotically information-orthogonal to $\beta_{bm:D}$ when paths are randomized independently of the biomarker (derived, Section 4.6).

What the illustration shows. On stored stress-test trials (Section 6; illustration, not a size study; 1,000 pooled null trials, MCSE about 0.007), F1 at the fixed 0.5 week rejected 0.061, F1A 0.074 and F1A0 0.067 on the same trials: F1A0 removes about half of F1A's excess, not all. The procedures that do not commit to one selected value held size: AIC-weight averaging 0.053 (0.049 with null-model weights), cross-fitting with a Cauchy combination 0.050, and the maximum $|t|$ over the grid with bootstrap calibration 0.048. A rule that keeps 0.5 week unless the null-model profile rejects it (F1G, post hoc) behaved as F1 (0.063) and switched in 1% to 15% of trials. Power in the base alternative cell was 0.400 for F1 and F1G, 0.39 for averaging and the bootstrap max-$|t|$, 0.29 for the Cauchy cross-fit and 0.21 for a single split (verified).

Provisional recommendation. Keep F1 at the half-life fixed in advance as the primary analysis, guarded by the data: switch to the null-model ML choice only when the trial's profile likelihood rejects the fixed value (F1G). If the half-life must always be estimated, use AIC-weight averaging over the grid. Report the half-life estimate and its profile-likelihood interval in every case. Confirm the choice, with F1AB and the bootstrap max-$|t|$ test as the calibration references, in a stress-test extension of 37,000 trials under the refined process (about 9 hours on 8 workers; Section 7.3). All of this is provisional.

2. Scope and relation to `docs/56`

`docs/56` states the recommended model F1, its expectancy terms, the pre-exposure guard on $D_{bc,it}$, its covariance and its CR2 inference, and summarizes the half-life issue in its Section 6. This note is the full treatment of that issue and does not repeat the model's justification. It concerns only how $t^a_{1/2}$ is chosen and how the test of $\beta_{bm:D}$ accounts for the choice. It does not revisit the expectancy terms, the covariance structure or the robustness of F1 to a prognostic biomarker, except where a half-life procedure interacts with them. All numerical work is on the Hybrid design at $N = 70$.

3. The problem

3.1 Where the half-life enters

F1 (`docs/56`, Section 2) is the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B^c_i + \beta_T w_t + \beta_D D_{bc,it}(h) + \beta_{bm:D}\, B^c_i D_{bc,it}(h) + \beta_\eta \eta_t + \beta_{bm:\eta}\, B^c_i \eta_t + u_i + \varepsilon_{it},$$

with $u_i \sim \mathcal{N}(0, \sigma^2_u)$, ε_{it} continuous-time AR(1) in weeks, the baseline as a response row at $t = 0$ with $D_{bc,it} = \eta_t = 0$, and $H_0 : \beta_{bm:D} = 0$ tested with the CR2 standard error and Satterthwaite degrees of freedom [3, 4]. The half-life enters through one regressor, $D_{bc,it}(h)$, in two places: the drug main effect and the interaction. In R, as in the stress-test driver (`11-strawman-stress-test.R`, lines 203 to 206 and 239; inspected):

```
dbc <- function(Db, tsd, h) {
  if (h <= 0) return(as.numeric(Db))
  ifelse(Db, 1, ifelse(tsd > 0, 0.5^(tsd / h), 0))
```

```
}
F_F1 <- Sx ~ bmc + t + X + bmc:X + De + bmc:De       # X = dbc(Db, tsd, h)
```

At $h = 0$ the regressor is the binary D_{it}; as $h \to \infty$ it becomes 1 at every visit after first exposure. In the Hybrid design the half-life changes $D_{bc,it}$ only at the off-drug visits, all of which follow exposure: weeks 11, 12 and 20 on path 1; 11, 12 and 16 on path 2; 10, 11, 12 and 20 on path 3; and 10, 11, 12 and 16 on path 4 (inspected, the `designs` list of `hendrickson-problems/04-power-simulation.R`, which the driver reads). The fits use `nlme::lme` with `corCAR1` [2].

3.2 The analysis half-life is a working parameter

$t^a_{1/2}$ and $t_{1/2}$ are different objects (argued). In the data-generating process the drug-response mean rises along a modified Gompertz curve while on drug and decays from its last on-drug value off drug [1]; the analysis replaces the whole on-drug curve with a constant β_D and the off-drug decay with $\beta_D 2^{-t_{sd}/h}$. The value of h that best fits a given trial therefore reflects the on-drug shape, the visit schedule and the correlation structure as well as the pharmacology. In the stored stress test, which used the published recursive carryover that decays faster than its stated half-life (`docs/53`), AIC chose 1 or 2 weeks in 85% of null trials of the compound-symmetry cell although the stated half-life was 0.5 week (verified, Table 3). A data-chosen $t^a_{1/2}$ is an estimate of the best-fitting exposure regressor, not of the drug's carryover half-life, and should be reported as such.

3.3 Why a fixed half-life keeps its size even when it is wrong

Under the null with a purely predictive biomarker ($c_{bm} = 0$), the biomarker is unrelated to how the outcome depends on exposure. For any regressor $D_{bc,it}(h)$ fixed before the outcome is seen, the true mean has no component along $B^c_i D_{bc,it}(h)$ that is not already absorbed by the other terms, so the probability limit of $\hat{\beta}_{bm:D}$ is 0 for every h (derived; compendium, eq:bp-plim, and `docs/56`, Section 6.2). A wrong h mis-scales the alternative: it moves the estimand and costs power, but it does not create false positives. The stress test agrees (quoted, `docs/47`, Section 3, and `docs/56`, Section 6.2): F1 at 0.5 week rejected 0.049 when the true half-life was 0 and 0.061 when it was 1 week. With a prognostic biomarker the leakage terms of `docs/56` depend mildly on h; no stored cell combined a prognostic pathway with a wrong half-life.

3.4 Why choosing the half-life by AIC inflates size

The identity. Every candidate model has the same parameters, so AIC differences are likelihood differences (compendium, eq:invd-aic). Writing $\ell_1(h)$ and $\ell_0(h)$ for the ML log-likelihoods with and without the interaction,

$$-\tfrac{1}{2}\mathrm{AIC}_1(h) + \mathrm{const} = \ell_1(h) = \ell_0(h) + \tfrac{1}{2}\mathrm{LR}(h), \qquad \mathrm{LR}(h) \approx T(h)^2$$

(derived; the approximation is the usual equivalence of the likelihood ratio and the squared Wald statistic, here with a CR2 rather than a model-based standard error). In the example trial of Section 5, $\mathrm{LR}(0) = 0.678$ against $T(0)^2 = 0.641$, and $\mathrm{LR}(2) = 0.0016$ against $T(2)^2 = 0.0016$ (verified). F1A therefore maximizes

$$\ell_0(h) + \tfrac{1}{2}T(h)^2 \quad \text{approximately:}$$

the drug-trajectory information in ℓ_0 plus a bonus for a large interaction statistic. It then compares $T(\hat{h}_1)$ with the reference distribution of a test at a fixed h, the general post-selection problem [16]. Two limiting cases bracket its behavior (derived from the display):

- ℓ_0 **steep.** If the drug main effect pins the half-life, so that differences in ℓ_0 across $\mathcal{T}_a$ are large compared with the variation of $T(h)^2/2$, then $\hat{h}_1 = \hat{h}_0$ in most trials and F1A behaves like F1A0.
- ℓ_0 **flat.** If the candidate regressors are nearly indistinguishable through the drug main effect, then $\hat{h}_1 \approx \arg\max_h |T(h)|$ and F1A becomes the max-$|t|$ test of Section 4.7 referred to a single-t critical value. Its size is then $\Pr(\max_h |T(h)| > t_{0.975})$, which exceeds α by an amount set by how correlated the five $T(h)$ are.

The inflation is mild because the $T(h)$ are strongly correlated across h (in the example trial they move from -0.80 to -0.04 over the grid, smoothly; Figure 1), and worst where ℓ_0 is flat, that is, where the half-life is weakly identified (argued).

The stored evidence. Table 2 compares F1A with F1 on the same null trials, by the half-life F1A selected.

Table 2: Null rejection rates of F1 (fixed 0.5 week) and F1A on the same trials of the 11 non-prognostic null cells, by the half-life F1A selected (verified from `replicates-full.rds` and `replicates-full-extra2.rds`; stored setting)

Selected $t^a_{1/2}$ (weeks)	Trials	F1 at 0.5	F1A
0	3,148	0.051	0.053
0.25	1,656	0.048	0.051
0.5	2,934	0.051	0.051
1	1,967	0.056	0.064
2	1,295	0.046	0.079
all	11,000	0.051	0.057

When F1A keeps 0.5 week it is F1. When it moves away, and most of all to 2 weeks, it rejects more often than F1 on the same trials; F1's own rate in those trials stays near 0.05, so the excess comes from the selection, not from the trials (verified). `docs/56`, Section 6.1, reports the same comparison with the mean $|t|$.

3.5 How much the data say about the half-life

Table 3: Share of null trials in which F1A's AIC chose each analysis half-life (weeks), selected cells, 1,000 trials each, and pooled over all cells and hypotheses (verified; stored setting)

Null cell	True $t_{1/2}$	0	0.25	0.5	1	2
base (AR(1), $\rho = 0.7$)	0.5	0.232	0.230	0.379	0.121	0.038
$t_{1/2} = 0$	0	0.870	0.082	0.044	0.003	0.001
$t_{1/2} = 1$ week	1	0.012	0.027	0.194	0.417	0.350
compound symmetry, $\rho_{cs} = 0.8$	0.5	0.019	0.025	0.110	0.382	0.464
$N = 35$	0.5	0.317	0.147	0.286	0.146	0.104
$N = 150$	0.5	0.138	0.286	0.479	0.092	0.005
Weibull decay, shape 2	0.5	0.710	0.167	0.107	0.012	0.004
all 41 cells		0.236	0.163	0.292	0.190	0.120

Three readings (verified unless stated):

- **The pooled split** of `docs/47` (24, 16, 29, 19 and 12%) mixes cells with different truths and should not be read as "AIC cannot find the half-life".
- **The likelihood separates "little carryover" from "a week or more"**, not 0.25 from 0.5 or 1: 87% at 0 when the truth is 0, 77% at 1 or 2 weeks when it is 1 week, but the true 0.5 only 38% of the time in the base cell. At $N = 150$ the share at 0.5 rises to 0.48.
- **The choice depends on things other than the half-life**: the correlation structure (compound symmetry pushes it to 1 or 2 weeks) and the decay shape (a Weibull shape-2 decay with the same half-life is chosen as 0 in 71% of trials). This is the working-parameter point of Section 3.2 (argued).

4. Methods

Each method is described by its idea, the intuition, an exact procedure, a validity argument with its conditions, its cost, what it reports, and its status. *Evaluated* means run at the stress-test scale (`docs/47`); *illustrated* means run only in the small Monte Carlo of Section 6; *proposed* means not run. Costs are on the machine used for `docs/47` (a 4 performance plus 4 efficiency core machine): one F1 fit with CR2 takes a fraction of a second, and the five-point grid with and without the interaction (15 fits and 5 CR2 computations) about 1.1 seconds on one worker (verified).

The code below uses the helpers of `analysis/scripts/data-chosen-half-life/57-common.R`: `with_x(d, h)` adds `X` $= D_{bc,it}(h)$, `fit_lme(f, d, method)` fits F1-type models with a random intercept and `corCAR1`, and `cr2t(fit)` returns the CR2 estimate, standard error, t, Satterthwaite df and p for `bmc:X`.

```
FM1 <- Sx ~ bmc + t + X + bmc:X + De + bmc:De   # F1
FM0 <- Sx ~ bmc + t + X + De + bmc:De           # F1 without the interaction
G5  <- c(0, 0.25, 0.5, 1, 2)                    # the grid T_a, weeks
ll  <- function(f, d, h) as.numeric(logLik(fit_lme(f, with_x(d, h), 'ML')))
```

4.1 Method 1: fixed in advance, with a sensitivity grid

Idea. Set $t^a_{1/2}$ from pharmacology before the data are seen (0.5 week in the project's PTSD and prazosin context), test at that value, and report F1 over the grid as a sensitivity analysis.

Intuition. A regressor chosen without the outcome cannot borrow the outcome's noise.

Procedure.

```
fit <- fit_lme(FM1, with_x(d, 0.5))             # REML
cr2t(fit)                                        # primary test
sapply(G5, function(h) cr2t(fit_lme(FM1, with_x(d, h)))['t'])   # sensitivity
```

Validity. Exact in the sense that matters here: no selection occurs, so the size is that of the CR2 test at a fixed regressor, whatever the true half-life (Section 3.3). Conditions: those of F1 itself (`docs/56`). The sensitivity grid is descriptive; it must not be used to pick the reported value.

Cost. One fit per trial; five for the sensitivity grid.

Reports. $\hat{\beta}_{bm:D}$ at the stated $t^a_{1/2}$ with its CR2 interval, and the five sensitivity t-statistics.

Why it is not enough. If the trial's profile likelihood excludes the stated value (as it would exclude 1 week in the example trial of Section 5), a reader can fairly ask why the analysis uses a regressor the data contradict. A fixed value may also lose power when it is far from the best-fitting one: with $t^a_{1/2} = 0.5$, F1's size-adjusted power at $c_{bm} = 0.30$ was 0.496 when the true half-life was 0 and 0.272 when it was 1 week (quoted, `docs/47`, Section 6). How much of that fall is due to the mismatch, rather than to the longer off-drug fading of the coupling itself, has not been separated.

Status. Evaluated (`docs/47`): size 0.051 pooled, at most 0.062 in a cell, stored setting.

4.2 Method 2: selection blind to the interaction (F1A0)

Idea. Choose $t^a_{1/2}$ by AIC (equivalently ML, Section 3.4) on F1 *without* the interaction term; refit F1 with the interaction at the chosen value by REML; test with CR2.

Intuition. The half-life is identified, as far as it is identified at all, by the drug-response trajectory: how far the outcome falls back at the off-drug visits. That information sits in the drug main effect. Removing the interaction from the selection model removes the $\frac{1}{2}T(h)^2$ bonus of Section 3.4.

Procedure.

```
ll0 <- sapply(G5, function(h) ll(FM0, d, h))    # ML, no interaction
h0  <- G5[which.max(ll0)]
cr2t(fit_lme(FM1, with_x(d, h0)))               # REML refit, CR2 test
```

Validity (derived to first order, with conditions). Two arguments, for the two regimes of Section 3.4.

1. *Weak identification: selection on the outcome alone.* Suppose, as in every non-prognostic null cell, that the biomarker is independent of the whole outcome vector Y and of the path assignment. Consider first a selection rule that uses Y and the design but no biomarker term, $\hat{h} = g(Y)$. Conditional on Y, $\hat{h}$ is fixed and the biomarker values keep their marginal distribution. To first order the CR2 statistic at any fixed h is a self-normalized sum,

$$T(h) \approx \frac{\sum_i B^c_i \, a_i(h, Y)}{\left\{ \sum_i (B^c_i)^2 \, a_i(h, Y)^2 \right\}^{1/2}},$$

   where B^c_i is the centered biomarker and $a_i(h, Y)$ is participant i's contribution to the exposure contrast after the other terms are partialled out (treating, to first order, the partialling of the biomarker terms B^c_i and $B^c_i \eta_t$ as fixed). Given Y, the B^c_i are exchangeable and mean zero, so $T(h)$ is approximately standard normal for every h and every Y for which no single participant dominates the weights. The rejection probability given Y is then approximately α whatever $g(Y)$ selects, and so is the unconditional size (derived to first order). The argument does not use identification of the half-life at all; it covers the flat case, where F1A fails.
2. *Strong identification: orthogonality.* When the drug main effect identifies the half-life, $\hat{h}_0$ is a consistent estimator of the best-fitting value, and the half-life is asymptotically information-orthogonal to $\beta_{bm:D}$ under the null when paths are randomized independently of the biomarker (derived, Section 4.6). Estimating an orthogonal nuisance parameter does not change the first-order distribution of the test.

F1A0 as specified keeps B^c_i and $B^c_i \eta_t$ in the selection model, so $\hat{h}_0$ depends on the biomarker through those terms. Their contribution to $\ell_0(h)$ changes with h only through the partialling of $D_{bc,it}(h)$, a second-order effect (argued). A stricter variant drops every biomarker term from the selection model and satisfies argument 1 as stated.

What would make it fail (argued):

- *A prognostic biomarker.* Then B^c_i is not independent of Y, and argument 1 no longer applies. F1's protection against prognostic leakage is approximate and depends mildly on h (`docs/56`), so a selection that moves h with the outcome could interact with the leakage. The illustration includes a prognostic null cell (Section 6).
- *Informative dropout related to the biomarker,* or allocation that depends on the biomarker, which break the independence and the orthogonality.
- *Very small N,* where the normal approximation to the self-normalized sum is poor; this affects the fixed-h test too, which is why CR2 with Satterthwaite degrees of freedom is used.

- *Under the alternative,* the selection model omits a true term and can choose h to absorb part of the interaction through the main effect. That costs power, not size.

Cost. Five ML fits and one REML fit with CR2 per trial, about 0.5 seconds.

Reports. $\hat{h}_0$, and $\hat{\beta}_{bm:D}$ with its CR2 interval at $\hat{h}_0$.

A guarded variant (F1G; post hoc). Keep the value fixed in advance, $t^a_{1/2} = 0.5$, unless the null-model profile rejects it at level 0.05, $2\{\max_h \ell_0(h) - \ell_0(0.5)\} > \chi^2_{1,0.95}$, and only then use $\hat{h}_0$:

```
dev05 <- 2 * (max(ll0) - ll0[G5 == 0.5])
h_use <- if (dev05 > qchisq(0.95, 1)) G5[which.max(ll0)] else 0.5
```

It answers the face-validity objection directly (the analysis never uses a half-life that its own data reject) while selecting in only the few trials where the default is contradicted, so whatever selection effect F1A0 has is confined to those trials (argued). It was defined after the illustration's results were seen.

Status. Illustrated (Section 6): 0.067 pooled null against F1's 0.061 on the same trials; F1G 0.063. Not evaluated at full scale. `docs/56`, Section 6.1, found that on 2×50 null trials F1A0 picked the same grid value as F1A in 86% and 88% of trials (quoted); here it did in 87% and 89% (verified, Table 8).

4.3 Method 3: sample splitting and cross-fitting

Idea. Choose the half-life on one random half of the participants and test on the other half, so that the test never sees the data that chose its regressor [8]. Cross-fitting swaps the halves and combines the two tests, recovering most of the lost information [9].

Intuition. Participants are independent. Once the half-life is a function of half A only, it is a constant as far as half B is concerned.

Procedure. Split within path, so that both halves keep the design's balance; choose with the F1A0 criterion (any criterion is valid here, including F1A's); test with F1 at the other half's choice.

```
ids <- unique(data.table(ptID = d$ptID, path = path_of(d)))
ids[, half := sample(rep(1:2, length.out = .N)), by = path]
dA <- d[ptID %in% ids[half == 1, ptID]]; dB <- d[ptID %in% ids[half == 2, ptID]]
choose <- function(dd) G5[which.max(sapply(G5, function(h) ll(FM0, dd, h)))]
hA <- choose(dA); hB <- choose(dB)
tB <- cr2t(fit_lme(FM1, with_x(dB, hA)))   # single split: test on B at hA
tA <- cr2t(fit_lme(FM1, with_x(dA, hB)))   # swapped
## cross-fit by averaging, with a Welch-Satterthwaite df
est <- (tA['est'] + tB['est']) / 2
se  <- sqrt(tA['se']^2 + tB['se']^2) / 2
df  <- (tA['se']^2 + tB['se']^2)^2 / (tA['se']^4 / tA['df'] + tB['se']^4 / tB['df'])
p_avg <- 2 * pt(-abs(est / se), df)
## or by the Cauchy combination of the two p-values [10]
p_cct <- 0.5 - atan(0.5 * tan((0.5 - tA['p']) * pi) + 0.5 * tan((0.5 - tB['p']) * pi)) / pi
```

Validity.

- *Single split: exact with respect to the selection* (derived). Conditional on half A, h_A is fixed and half B is independent of half A, so the test on B is a fixed-h test. Its size is the fixed-h size of F1 at $N/2$; no further condition is needed.

- *Cross-fit average: approximate* (derived to first order). Under the null, $E[\hat{\beta}_A \mid \text{half B}] \approx 0$ because given half B the regressor for half A is fixed, so the two half estimates are approximately uncorrelated and the variance of their average is approximately $(\mathrm{SE}_A^2 + \mathrm{SE}_B^2)/4$. They are not independent, since each half's data choose the other's regressor.
- *Cauchy combination: approximate.* Valid in the upper tail under arbitrary dependence as $\alpha \to 0$ [10]; at $\alpha = 0.05$ it is close to, and slightly below, nominal for two moderately dependent p-values (argued). Bonferroni, $2\min(p_A, p_B)$, is exact and conservative.

Cost. Ten ML fits on half the data and two REML fits with CR2: about the cost of F1A0. The cost that matters is power. A single split tests on $N/2$, so the noncentrality falls by a factor of about $\sqrt{2}$; the cross-fit average uses all participants but each at a regressor chosen on half the data (argued; Section 6 measures it).

Reports. The two choices h_A and h_B, the cross-fit estimate and its interval or the combined p-value. The result depends on the random split; the split seed should be fixed in advance and reported, or the analysis repeated over many splits and summarized (for example by the median p-value, which needs its own calibration).

Status. Illustrated (Section 6): single split 0.044 pooled null, power 0.208; cross-fit average 0.061 and 0.372; Cauchy 0.050 and 0.292.

4.4 Method 4: reporting the half-life

Idea. Whatever the test uses, report what the trial says about the half-life: the ML estimate under the null model, the profile likelihood over a fine grid or a continuous range, and a profile-likelihood interval.

Intuition. The profile shows directly which half-lives the data contradict, which is the face-validity question.

Procedure.

```
fine <- c(0, 0.02, 0.05, seq(0.1, 1, 0.05), seq(1.1, 3, 0.1), 3.5, 4, 5, 6, 8, 10, 15, 20, 50)
prof <- sapply(fine, function(h) ll(FM0, d, h))
opt  <- optimize(function(lh) ll(FM0, d, exp(lh)), log(c(0.01, 100)), maximum = TRUE)
lmax <- max(prof, opt$objective)
ci   <- range(fine[2 * (lmax - prof) <= qchisq(0.95, 1)])
```

Validity. The interval is the set of half-lives not rejected by a likelihood-ratio test at level 0.05 with a χ^2_1 reference. That reference is not exact here (derived from standard theory): the parameter has a boundary at 0, where the binary indicator lies, and a second one at ∞, where $D_{bc,it}$ becomes an ever-exposed indicator [11]; and when the profile is flat, the half-life is weakly identified and the χ^2 approximation fails. The interval should be read as a description of the likelihood, not as an interval with known coverage. Its coverage of the best-fitting working value has not been evaluated. It is an interval for the working parameter (Section 3.2), not for the pharmacological half-life.

Cost. On a 52-point grid, two ML fits per point, plus a REML refit and CR2 for the t-profile: 11 seconds per trial (verified, Section 5); once per trial, not per resample.

Reports. $\hat{t}^a_{1/2}$, the interval, and the deviance at 0 and at long half-lives. The honest statement for a Hybrid trial is that its data usually separate a short half-life from one of a week or more, but not 0 from 0.25 or 0.5 week (Sections 3.5 and 5).

Status. Illustrated on one trial (Section 5); coverage not evaluated. Recommended for reporting in every case.

4.5 Method 5: parametric bootstrap of the whole procedure (F1AB)

Idea. Calibrate the select-then-test statistic by simulating it under a fitted null model that reproduces the trial's design, its estimated drug trajectory and its estimated covariance, rerunning the selection in every resample.

Intuition. F1A's flaw is that its reference distribution ignores the selection. A bootstrap that repeats the selection includes it.

Procedure.

1. Fit the null model FM0 with its own selected half-life $\hat{h}_0$ (ML choice, REML refit), giving $\hat{\xi}$, $\hat{\sigma}_u$, $\hat{\sigma}_\varepsilon$ and the CAR(1) parameter $\hat{\rho}_a$.
2. For $m = 1, \ldots, M$: simulate $Y_i^* = W_i(\hat{h}_0)\hat{\xi} + u_i^* + \varepsilon_i^*$ for every participant, with the observed visits, biomarker values and paths; rerun the whole procedure (selection and test) on Y^* and record $|T_m^*|$.
3. Report $p^* = (1 + \#\{m : |T_m^*| \geq |T_{\text{obs}}|\})/(M+1)$.

```
fit0 <- fit_lme(FM0, with_x(d, h0))                    # REML null fit at h0
sim_null <- function(fit, d, h0) {
  dd  <- with_x(d, h0)
  mu  <- drop(model.matrix(FM0[-2], dd) %*% nlme::fixef(fit))
  tau <- as.numeric(nlme::VarCorr(fit)['(Intercept)', 'StdDev'])
  phi <- as.numeric(coef(fit$modelStruct$corStruct, unconstrained = FALSE))
  y <- mu
  for (idx in split(seq_len(nrow(dd)), dd$ptID)) {
    tt <- dd$t[idx]
    S  <- tau^2 + fit$sigma^2 * phi^abs(outer(tt, tt, '-'))
    y[idx] <- mu[idx] + drop(crossprod(chol(S), rnorm(length(idx))))
  }
  ds <- copy(d); ds[, Sx := y]; ds
}
tstar <- replicate(M, abs(stat_F1A(sim_null(fit0, d, h0))))   # stat_F1A: select, refit, CR2 t
p_boot <- (1 + sum(tstar >= abs(t_obs))) / (M + 1)
```

Validity. By construction, up to the fit of the null model (argued, with the usual parametric-bootstrap conditions). The bootstrap distribution of the selected statistic is computed under a null model with the estimated parameters; if that model describes the null data well, p^* is approximately uniform. It can fail when the null model is misspecified in ways that matter for the statistic's distribution: a residual covariance other than random intercept plus CAR(1), on-drug means that a constant β_D fits poorly, or a prognostic biomarker whose association with Y the null model reproduces only through B_i^c and $B_i^c\eta_t$. Because the resamples keep the observed biomarker values, paths and visits, the bootstrap conditions on the design. The bootstrap is calibrated against the selection under the fitted null model; it does not repair a misspecified analysis model.

Choice of M. For a test at 0.05, $M = 199$ or $M = 999$ (so that $(M+1)\alpha$ is an integer). The Monte Carlo standard error of p^* near 0.05 is about $\sqrt{0.05 \times 0.95/M}$, 0.015 at $M = 199$ and 0.007 at $M = 999$ (derived).

Cost. Each resample costs one run of the procedure: about 1.1 seconds for the grid with and without the interaction. For one trial, $M = 199$ took 128 seconds on 7 workers (verified, Section 5); $M = 999$ would take about 11 minutes. In a simulation study, a full bootstrap per replicate multiplies the cost by M. The warp-speed method of Giacomini, Politis and White [5] draws one resample per Monte Carlo replicate and uses the pooled resampled statistics of a cell as the bootstrap distribution, which estimates the rejection rate of the bootstrap test at about twice the cost of the plain procedure. Its justification requires the bootstrap distribution to vary little across replicates of a cell [5] (assumed here); its rejection rate has a Monte Carlo standard error somewhat larger than the binomial one reported below.

Reports. p^* for the chosen statistic (F1A, F1A0 or max-$|t|$), the bootstrap critical value, and $\hat{h}_0$.

Status. Illustrated by the warp-speed method (Section 6: F1AB 0.062 pooled null, F1's own rate on the same trials being 0.061; power 0.364) and on one trial with $M = 199$ (Section 5).

4.6 Method 6: joint estimation of the half-life as a continuous parameter

Idea. Treat $\omega = \log t^a_{1/2}$ as a parameter of a nonlinear mixed model in which $D_{bc,it}(e^\omega)$ enters both the main effect and the interaction, estimate it with the other parameters, and test $\beta_{bm:D}$ with a standard error that includes the uncertainty in $\hat{\omega}$.

Intuition. This is the textbook answer: estimate every unknown, and let the information matrix decide how much the estimation of ω costs the test.

Procedure (sketch; not implemented).

- *Estimation:* either `nlme::nlme()` with the mean function $\beta_0+\beta_B B^c_i+\beta_T w_t+(\beta_D+\beta_{bm:D}B^c_i)\, D_{bc,it}(e^\omega)+\beta_\eta\eta_t + \beta_{bm:\eta}B^c_i\eta_t$, a random intercept and `corCAR1`, or profile $\hat{\omega} = \arg\max_\omega \ell_1(e^\omega)$ by `optimize()` and fit the linear model at $\hat{\omega}$.
- *Inference:* a Wald test with the sandwich variance of the stacked estimating equations for (β, ω), in which the derivative $\partial D_{bc,it}/\partial\omega = D_{bc,it}\, \ln 2\; t_{sd}\, e^{-\omega}$ (off drug after exposure; 0 otherwise) enters the bread, or a score test of $\beta_{bm:D} = 0$ at the null fit $(\hat{\omega}_0, \hat{\xi}_0)$ with the efficient score $U_\beta - I_{\beta\omega}I^{-1}_{\omega\omega}U_\omega$.

Validity and the orthogonality result. Under the null, the cross-information between $\beta_{bm:D}$ and ω is

$$I_{\beta\omega} = \beta_D \sum_i B^c_i\, D_{bc,i}(e^\omega)^\top V_i^{-1}\, \frac{\partial D_{bc,i}}{\partial\omega},$$

plus terms with mean-zero residuals (derived). The vectors $D_{bc,i}$ and $\partial D_{bc,i}/\partial\omega$ are the same for all participants on the same path, so the sum is $\beta_D \sum_q c_q \sum_{i\in q} B^c_i$ over paths q, and is $O_p(N^{1/2})$ when paths are assigned independently of the biomarker, against $O(N)$ for $I_{\beta\beta}$ and $I_{\omega\omega}$ (derived). The half-life is therefore asymptotically orthogonal to the interaction under the null: its estimation does not change the first-order null distribution of the score test, and F1A0 is that score test with ω restricted to a grid. The orthogonality needs three conditions: randomization independent of the biomarker, β_D and $I_{\omega\omega}$ bounded away from zero (a drug effect that identifies the half-life), and ω in the interior. The last two fail in practice: $I_{\omega\omega}$ is small in a Hybrid trial (Section 3.5), the ML estimate is often on the boundary at 0 (it is in the example trial, Section 5), and then the asymptotic argument says nothing [11]. The Wald test at the joint estimate also inherits F1A's problem when ω is weakly identified, since the joint estimate of ω then responds to the interaction's noise.

Small samples. No CR2 analog exists for a nonlinear mean parameter; a CR2-type correction would have to be developed and checked (argued).

Cost. One `nlme` fit per trial, typically several times an `lme` fit, with convergence failures near the boundaries; or a profile over ω (about 25 one-dimensional evaluations) plus a sandwich.

Reports. $\hat{t}^a_{1/2}$ with a standard error, and the interaction test adjusted for its estimation.

Status. Proposed. Its main contribution here is the orthogonality argument, which supports F1A0.

4.7 Method 7: the maximum statistic over the grid

Idea. Compute $T(h)$ at every $h \in \mathcal{T}_a$ and test with $\max_h |T(h)|$, calibrated by its joint null distribution.

Intuition. Instead of guessing the half-life, ask whether there is an interaction at *some* half-life on the grid, and pay for the multiplicity explicitly. This is the problem of a nuisance parameter present only under the alternative studied by Davies [6, 7] and Hansen [12]: if β_D were 0, h would appear only in the interaction term and be unidentified under the null.

Procedure.

```
Th <- sapply(G5, function(h) cr2t(fit_lme(FM1, with_x(d, h)))['t'])
Tmax <- max(abs(Th))
p_bonf <- min(1, length(G5) * min(2 * pt(-abs(Th), df)))   # Bonferroni: exact, conservative
## calibrated: compare Tmax with the max over the grid in null-model resamples (Section 4.5)
## or with the max of a multivariate normal with the correlation of the T(h)
```

The correlation of the $T(h)$ can be estimated from the stacked participant-level influence contributions of the five fits (a cluster-robust covariance of the five interaction estimates), and the critical value read from the maximum of a multivariate normal or t with that correlation (`mvtnorm`), or taken from the null-model bootstrap.

Validity. Bonferroni is exact with respect to the multiplicity if each $T(h)$ is a valid fixed-h test, and conservative because the $T(h)$ are strongly correlated. The bootstrap calibration is valid up to the fit of the null model (Section 4.5). The multivariate normal calibration is approximate (first order). Davies' bounds [6, 7] cover the continuous-h version.

What it tests. The same null hypothesis as F1 (no interaction, for any exposure regressor), but against a different alternative: an interaction whose best-fitting half-life is unknown. The reported quantity is a p-value and the half-life at which $|T|$ is largest; there is no single estimand, which weakens its face validity for a reader who wants an effect size.

Cost. Five REML fits with CR2 (about 1 second); bootstrap calibration adds M times that per trial.

Status. Illustrated (Section 6): Bonferroni 0.024 pooled null, power 0.220; warp-speed bootstrap 0.048 and 0.384.

4.8 Method 8: marginalizing over the half-life

Idea. Put a prior on $t^a_{1/2}$ informed by pharmacology, and base the decision on the posterior of $\beta_{bm:D}$ integrated over the half-life.

Intuition. The analyst does not know the half-life; averaging over the plausible values with weights given by prior and data is the coherent way to carry that uncertainty.

Procedure. Full Bayesian: a Stan model with the F1 mean function, $D_{bc,it}(e^{\omega})$ computed inside the model, a random intercept and CAR(1) residuals (not available directly in `brms`, which has no continuous-time AR(1) for irregular visits, so a custom Stan program is needed), a prior on ω (for example normal on the log scale centered at $\log 0.5$ with a standard deviation of 1), and weakly informative priors elsewhere. The decision rule might be: reject if the 95% central posterior interval of $\beta_{bm:D}$ excludes 0. A cheaper frequentist stand-in is model averaging over the grid with smoothed AIC weights (Buckland, Burnham and Augustin [13]):

```
w   <- exp(ll1 - max(ll1)); w <- w / sum(w)          # ll1: ML of F1 on the grid
est <- sum(w * est_h)                                # est_h, se_h: REML + CR2 at each h
se  <- sum(w * sqrt(se_h^2 + (est_h - est)^2))
```

Validity. Unknown. A posterior-interval decision rule has no guaranteed frequentist size; it must be calibrated by simulation, and the calibration depends on the prior. The weights use ℓ_1, which includes the

interaction, so averaging inherits a damped version of F1A's preference for half-lives at which $|T|$ is large (derived from Section 3.4). Bayesian model averaging is reviewed by Hoeting et al. [14].

Cost. An MCMC fit per trial, perhaps 1 to 5 minutes with a custom Stan program (assumed, not measured), which makes a 28,000-trial calibration expensive. The AIC-weight stand-in costs the grid fits only.

Reports. The posterior of $\beta_{bm:D}$ and of $t^a_{1/2}$; the stand-in reports a model-averaged estimate and interval.

Status. Proposed (Bayesian); the AIC-weight stand-in is illustrated (Section 6: 0.053 pooled null with ℓ_1 weights and 0.049 with ℓ_0 weights; power 0.388 and 0.356).

4.9 Permutation tests

Naive permutation of the biomarker is not valid for this null. Permuting B^c_i across participants generates the distribution of the statistic under the *strong* null that the biomarker is unrelated to the outcome in every respect. The hypothesis of interest is weaker: $\beta_{bm:D} = 0$ while B^c_i may be prognostic, with a main effect or an association that varies with expectancy. When the biomarker is prognostic, permutation destroys that association in the resamples but not in the observed data, so the observed statistic is compared with the wrong distribution (derived). For a studentized statistic with the prognostic terms in the model the damage may be small, but nothing guarantees it, and the test then answers a question nobody asked. When the biomarker is not prognostic the strong and weak nulls coincide and permutation within path is exact, but that is the case in which no help is needed.

What a valid version would require. A residual permutation in the manner of Freedman and Lane [15]: fit the null model FM0 (with its selected half-life), permute the participant-level residual vectors (all visits of a participant together, to keep the within-participant correlation) among participants *of the same path* (whose design rows are identical), add them to the fitted null values, rerun the selection and the test, and refer the observed statistic to the permutation distribution. This requires the residual vectors to be exchangeable within path, which holds only approximately (estimated residuals, dropout, phase-dependent variances), and it is valid only approximately, at about the cost of the parametric bootstrap.

Status. Proposed with conditions; not recommended over the parametric bootstrap, which needs fewer exchangeability assumptions and handles unequal path sizes and dropout directly.

5. Worked example 1: one null trial

5.1 Set-up

One trial was drawn from the base null cell of the stress test with the *refined* process of the driver (anchored carryover, randomized allocation; `--carryover anchored --allocation random`): $N = 70$, path sizes 14, 23, 13 and 20, AR(1) at $\rho = 0.7$, true $t_{1/2} = 0.5$ week, exponential decay, $c_{bm} = 0$, no dropout and no prognostic pathway (inspected, `57-example-trial.R`). The seed (5701) was fixed before the trial was seen, and the trial was not selected for any property.

5.2 The profile and the test across the half-life

Table 4: Example 1: ML log-likelihoods with and without the interaction, the likelihood-ratio statistic, and the REML + CR2 test of the interaction at each grid half-life (verified)

h (weeks)	$\ell_1(h)$	$\ell_0(h)$	LR(h)	$\hat{\beta}_{bm:D}$	CR2 SE	$T(h)$	df	p
0	−2591.658	−2591.997	0.678	−0.0602	0.0752	−0.801	26.0	0.430
0.25	−2591.899	−2592.201	0.605	−0.0589	0.0781	−0.753	25.9	0.458
0.5	−2593.667	−2593.852	0.371	−0.0510	0.0874	−0.584	26.0	0.564

h (weeks)	$\ell_1(h)$	$\ell_0(h)$	LR(h)	$\hat{\beta}_{bm:D}$	CR2 SE	$T(h)$	df	p
1	−2597.846	−2597.903	0.113	−0.0330	0.1022	−0.323	26.5	0.749
2	−2601.938	−2601.939	0.002	−0.0048	0.1202	−0.040	27.3	0.968

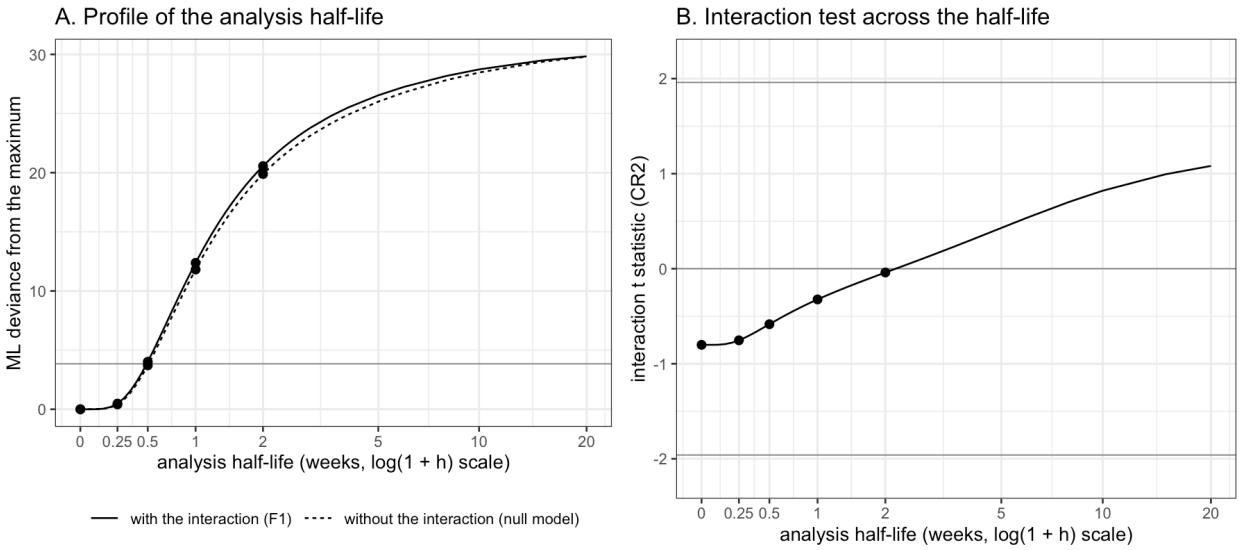


Figure 1: Example 1, one null Hybrid trial ($N = 70$, true $t_{1/2} = 0.5$ week, refined process). A: ML deviance $2\{\ell_{\max} - \ell(h)\}$ of F1 with the interaction (solid) and without it (dashed) over a fine grid of analysis half-lives; the horizontal line is $\chi^2_{1,0.95} = 3.84$, and points mark the grid $\mathcal{T}_a$. B: the CR2 t-statistic of the interaction across the same half-lives, with ± 1.96 marked.

Four observations (verified):

- **The profile is flat where it matters and steep elsewhere.** The null-model deviance is 0 at $h = 0$, 0.41 at 0.25 week and 3.71 at 0.5 week, then 11.8 at 1 week and 19.9 at 2 weeks, rising to 30.8 as $h \to \infty$. The trial cannot tell 0 from 0.5 week but excludes a week or more.
- **The ML estimate is on the boundary.** The continuous maximum over $[0.01, 100]$ weeks is at 0.024 week, the lower end of the range in practice, and $\ell_0(0)$ is at least as large: the binary indicator fits best, although the true half-life is 0.5 week. The profile-likelihood interval is $[0, 0.50]$ weeks for the null model and $[0, 0.45]$ for F1, so the true value lies at the upper edge of one and just outside the other; with a working parameter and a boundary estimate, neither interval has a known coverage (Section 4.4).
- **The two profiles are almost parallel.** LR(h) is below 0.7 at every h, so in this trial F1A and F1A0 make the same choice.
- **The interaction statistic changes smoothly and modestly with h:** from −0.80 at 0 to −0.04 at 2 weeks and +1.1 near 20 weeks.

5.3 What each method reports

Table 5: Example 1: what each procedure reports for the same null trial (verified)

Method	Half-life used (weeks)	$\hat{\beta}_{bm:D}$	Statistic	p
1. F1, fixed 0.5	0.5	−0.051	$t = -0.58$	0.564

Method	Half-life used (weeks)	$\hat{\beta}_{bm:D}$	Statistic	p
F1A (AIC with the interaction)	0	-0.060	$t = -0.80$	0.430
2. F1A0 (AIC without it)	0	-0.060	$t = -0.80$	0.430
3. single split (choose on A, test on B)	0	0.048	$t = 0.42$ (df 8.8)	0.684
3. cross-fit, average	0 and 0.25	-0.022	$t = -0.28$ (df 21.7)	0.781
3. cross-fit, Cauchy	0 and 0.25			0.544
4. profile interval	ML at 0; interval $[0, 0.50]$			
5. F1AB, bootstrap of F1A ($M = 199$)	0		$\lvert t\rvert = 0.80$; critical value 1.93	0.435
5. bootstrap of F1A0	0		critical value 1.92	0.430
5. bootstrap of F1 at 0.5	0.5		critical value 1.92	0.550
7. max-$\lvert t\rvert$, Bonferroni	0		$\max\lvert t\rvert = 0.80$	1.000
7. max-$\lvert t\rvert$, bootstrap ($M = 199$)	0		critical value 2.19	0.560
8. AIC-weight averaging	weighted mean 0.14	-0.059	$t = -0.76$	0.452

In this trial every method agrees that there is no evidence of an interaction, as it should. Three details are worth noting (verified):

- **The bootstrap critical values.** For a single fixed-h test the bootstrap gives 1.92, close to the t_{26} value 2.06 and below it (199 resamples, so the quantile has a standard error of roughly 0.13, derived from the binomial error of a 95% quantile). For F1A it gives 1.93, barely more: in this trial the selection adds little, because ℓ_0 is steep enough to pin the choice near 0. For the maximum over the grid it gives 2.19, the price of multiplicity across five strongly correlated statistics.
- **The single split** tests on 35 participants with only 8.8 Satterthwaite degrees of freedom; its estimate has the opposite sign and a standard error 1.5 times F1's. The cross-fit average recovers a standard error close to F1A0's (0.076 against 0.075).
- **Timing.** The fine profile (52 half-lives, three fits and one CR2 each) took 11 seconds; the five-point grid with the split 1.5 seconds; the bootstrap with $M = 199$ 128 seconds on 7 workers; the whole example 146 seconds.

6. Worked example 2: a small Monte Carlo illustration

6.1 Set-up

This is an illustration, not a replacement for the stress-test extension of Section 7.3. It uses the trials of the stored stress test exactly: the driver's data-generating functions, cells and seeds (chunks of 25 replicates, seed $20261007 + 1000 \times \text{cell} + \text{chunk}$) with `--carryover recursive --allocation fixed`, the setting of

the stored results, so that replicate r here is replicate r of `docs/47` (inspected, `57-null-illustration.R`). The pairing was checked: F1 and F1A reproduce the stored estimates exactly in every replicate (maximum absolute difference 0 in all 1,500). The split and the single bootstrap resample per replicate use a separate random stream, so they do not disturb the trials.

Four cells were run: the base null cell (cell 2, $n_{\text{sim}} = 500$), the null cell with true $t_{1/2} = 1$ week where F1A failed (cell 20, $n_{\text{sim}} = 500$), the null cell with a constant prognostic natural-course pathway $c_{bm,TV} = 0.3$ (cell 25, $n_{\text{sim}} = 250$), and the base alternative cell $c_{bm} = 0.30$ (cell 6, $n_{\text{sim}} = 250$). The replicate counts were cut from 1,000 and 500 to keep the run within the time budget (Section 6.4). Each replicate fits F1 at every grid value, the selection models, the split and cross-fit, and one null-model bootstrap resample on which the F1, F1A, F1A0 and max-$|t|$ procedures are rerun (warp-speed method [5]).

6.2 Results

Table 6: Example 2: rejection rates at nominal 0.05 (binomial MCSE in parentheses) on stored stress-test trials, stored setting (verified; the warp-speed MCSEs understate the error, and their pooled column is the mean of the two cell rates)

Method	Base null	Null, $t_{1/2} = 1$	Null, prognostic TV	Pooled null (base and $t_{1/2} = 1$)	Power, $c_{bm} = 0.30$
n_{sim}	500	500	250	1,000	250
F1, fixed 0.5	0.064 (0.011)	0.058 (0.010)	0.032 (0.011)	0.061 (0.008)	0.400 (0.031)
F1A	0.070 (0.011)	0.078 (0.012)	0.040 (0.012)	0.074 (0.008)	0.428 (0.031)
F1A0	0.064 (0.011)	0.070 (0.011)	0.040 (0.012)	0.067 (0.008)	0.396 (0.031)
F1G, guarded F1A0 (post hoc)	0.064 (0.011)	0.062 (0.011)	0.032 (0.011)	0.063 (0.008)	0.400 (0.031)
AIC-weight averaging, ℓ_1 weights	0.054 (0.010)	0.052 (0.010)	0.032 (0.011)	0.053 (0.007)	0.388 (0.031)
AIC-weight averaging, ℓ_0 weights (post hoc)	0.050 (0.010)	0.048 (0.010)	0.032 (0.011)	0.049 (0.007)	0.356 (0.030)
single split	0.046 (0.009)	0.042 (0.009)	0.044 (0.013)	0.044 (0.006)	0.208 (0.026)
cross-fit, average	0.056 (0.010)	0.066 (0.011)	0.032 (0.011)	0.061 (0.008)	0.372 (0.031)
cross-fit, Cauchy	0.050 (0.010)	0.050 (0.010)	0.032 (0.011)	0.050 (0.007)	0.292 (0.029)
max-$|t|$, Bonferroni	0.024 (0.007)	0.024 (0.007)	0.016 (0.008)	0.024 (0.005)	0.220 (0.026)
F1, warp-speed bootstrap	0.062 (0.011)	0.060 (0.011)	0.028 (0.010)	0.061	0.388 (0.031)
F1AB, warp-speed bootstrap of F1A	0.066 (0.011)	0.058 (0.010)	0.036 (0.012)	0.062	0.364 (0.030)
F1A0, warp-speed bootstrap	0.070 (0.011)	0.058 (0.010)	0.032 (0.011)	0.064	0.348 (0.030)

Method	Base null	Null, $t_{1/2} = 1$	Null, prognostic TV	Pooled null (base and $t_{1/2} = 1$)	Power, $c_{bm} = 0.30$
max-$\lvert t \rvert$, warp-speed bootstrap	0.052 (0.010)	0.044 (0.009)	0.040 (0.012)	0.048	0.384 (0.031)

Table 7: Example 2: warp-speed bootstrap critical values for $|T|$, the 95% quantile of the pooled resampled statistics of each cell (verified)

Procedure	Base cell	$t_{1/2} = 1$	Prognostic TV	$c_{bm} = 0.30$
F1, bootstrap	2.09	2.03	2.10	2.13
F1AB	2.11	2.24	2.13	2.21
F1A0, bootstrap	2.01	2.15	2.13	2.19
max-$\lvert t \rvert$, bootstrap	2.39	2.43	2.27	2.30

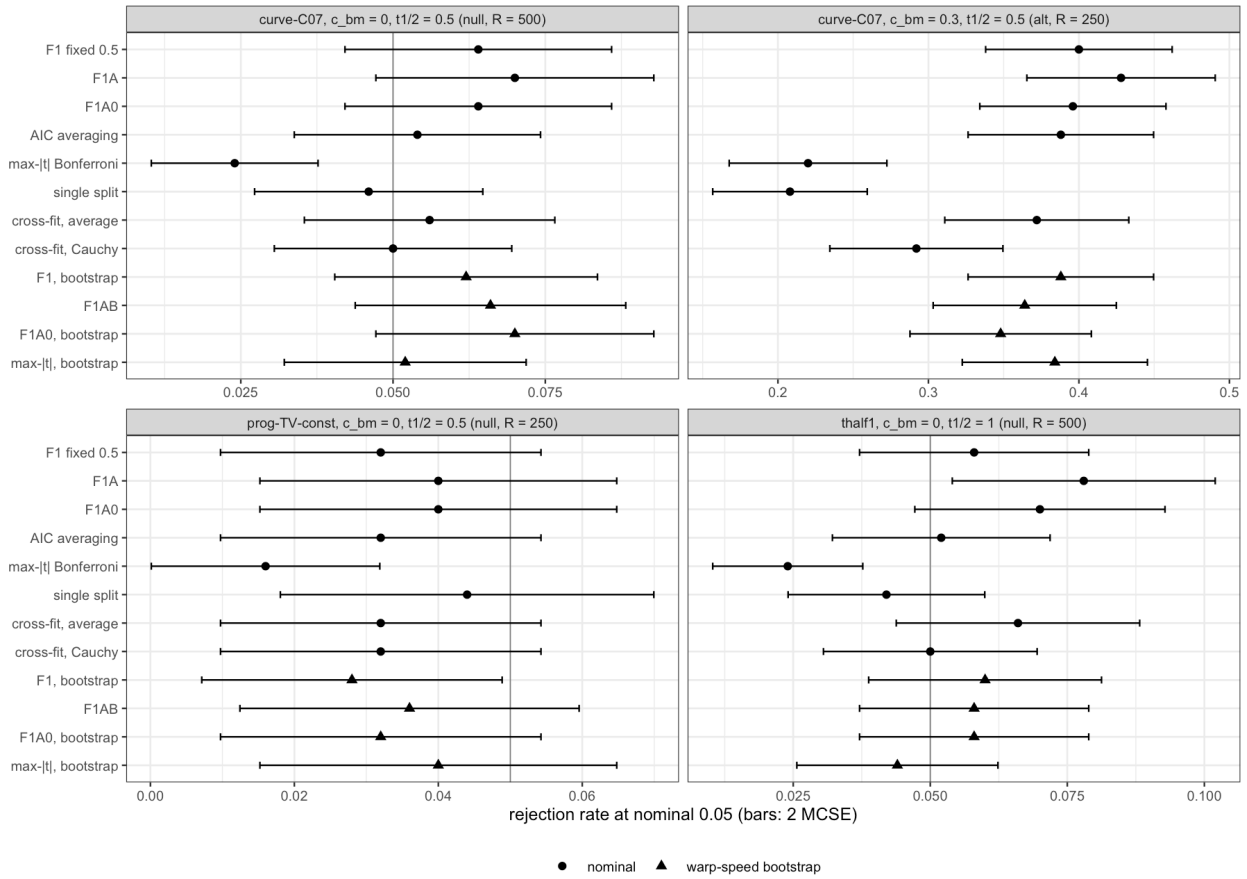


Figure 2: Example 2: rejection rates at nominal 0.05 with 2-MCSE bars, by procedure, in the four stress-test cells (stored setting). Circles: nominal reference distribution; triangles: warp-speed bootstrap calibration. The vertical line marks 0.05 in the null cells; the alternative cell (top right) shows power.

The post hoc rows were defined after the other results were seen and are computed from the same grid fits (`57-posthoc-rules.R`). F1G keeps $t^a_{1/2} = 0.5$ week unless the null-model profile rejects it ($2\{\max_h \ell_0(h) -$

$\ell_0(0.5)\} > 3.84$ on the grid), and only then uses $\hat{h}_0$; it switched in 1.0%, 15.0%, 2.0% and 3.6% of the trials of the four cells (verified).

6.3 Reading

All rates below are from Table 6 and the paired counts of `57-mc-discordance.csv` (verified); with 1,000 pooled null trials the MCSE of a rate near 0.05 is 0.007, so only differences of about 0.015 or more are resolved.

- **The pairing holds.** F1 and F1A reproduce the stored estimates and decisions in every replicate (maximum absolute difference 0), so every comparison is paired with `docs/47`. In these first 500 trials of the base cell F1 itself rejects 0.064 (0.062 over all 1,000 stored trials): the excess there is F1's, not the selection's.
- **F1A0 removes about half of F1A's excess, not all of it.** Pooled, F1A rejects 0.074, F1A0 0.067 and F1 0.061 on the same trials. In the $t_{1/2} = 1$ cell F1A0 rejected 10 trials that F1 did not, and F1 4 that F1A0 did not; F1A 12 and 2. F1A0's choices differ from F1A's in only 11 to 13% of null trials (Table 8), which is where the difference comes from. The first-order argument of Section 4.2 is therefore not borne out cleanly at $N = 70$, though the excess over F1 (0.006, paired) is within the Monte Carlo error.
- **The guard costs nothing measurable.** F1G is F1 except in the few trials whose data contradict 0.5 week; it rejected 2 null trials that F1 did not, out of 1,250, and had the same power (0.400).
- **Averaging over the grid holds size.** With ℓ_1 weights, 0.053 pooled; with ℓ_0 weights, 0.049. Both are slightly more conservative than F1 on the same trials (in the base cell F1 rejected 6 and 7 trials that they did not, against 1 and 0 the other way), because the averaged standard error includes the spread of the estimates across the grid. Their power cost against F1 is 0.012 and 0.044, within one and a half MCSEs.
- **Splitting is valid and expensive.** The single split holds size (0.044) and loses half the power (0.208 against 0.400). The cross-fit average recovers most of the power (0.372) but inherits a little of the selection effect (0.066 in the $t_{1/2} = 1$ cell); the Cauchy combination is at 0.050 with power 0.292.
- **The bootstrap calibrates the selection but not F1's own excess.** The warp-speed critical value of F1A exceeds that of F1 by 0.02 to 0.21, and F1AB's pooled rate (0.062) is F1's (0.061). The bootstrap of F1 at a fixed half-life does not bring F1 to 0.05 in the base cell (0.062), which is consistent with a null model (random intercept plus CAR(1), constant drug effect) that is lighter-tailed than the data-generating process (argued). The max-$|t|$ statistic with bootstrap calibration is at 0.048 pooled with power 0.384; its Bonferroni version is conservative (0.024) and loses half the power.
- **The prognostic cell** ($c_{bm,TV} = 0.3$) is below 0.05 for every procedure (0.016 to 0.044, $n_{\text{sim}} = 250$); no selection procedure reintroduced the prognostic leakage that F1 removes. This cell has the true half-life 0.5, so it does not test a prognostic biomarker combined with a wrong half-life.
- **Power is that of a cell where 0.5 week is right.** In the alternative cell the fixed value matches the truth, so no data-chosen procedure can gain; F1A's 0.428 is inflated by its size. The cells where a data-chosen half-life could gain power ($t_{1/2} = 0$ or 1 week under the alternative) were not run.

Table 8: Example 2: share of trials choosing each analysis half-life (weeks); F1A and F1A0 agreed in 86.6% and 89.4% of the trials of these two cells (verified)

Cell	Procedure	0	0.25	0.5	1	2
base null	F1A	0.234	0.234	0.394	0.110	0.028
base null	F1A0	0.218	0.242	0.416	0.096	0.028
base null	split (half A)	0.282	0.180	0.346	0.126	0.066
$t_{1/2} = 1$ null	F1A	0.008	0.036	0.192	0.416	0.348
$t_{1/2} = 1$ null	F1A0	0.004	0.028	0.218	0.386	0.364
$t_{1/2} = 1$ null	split (half A)	0.070	0.054	0.202	0.294	0.380

Cell	Procedure	0	0.25	0.5	1	2

6.4 Run time

The 1,500 replicates took 10,590 worker-seconds, 7.1 worker-seconds per replicate (one grid of 15 fits and 5 CR2 computations on the data, the split, and the same grid on one bootstrap resample), measured on 7 workers on the machine of `docs/47`; the final run took 23 minutes of wall time, after a first launch (with twice the replicates) that was stopped to stay within the time budget and whose completed chunks were reused (verified, `57-mc-timing.csv`). Example 1 took 146 seconds. The whole illustration therefore used about 30 minutes of wall time on 7 workers.

7. Recommendation

7.1 Decision table

"Size" and "power" quote the illustration of Section 6 (pooled null over the base and $t_{1/2} = 1$ cells, $n_{\text{sim}} =$ 1,000; power in the base alternative cell, $n_{\text{sim}} = 250$) unless marked otherwise. "Face validity" asks whether the analysis avoids a half-life the trial's data contradict and reports what the data say about it.

Table 9: Decision table for choosing the analysis half-life from the data (size and power from the illustration of Section 6, stored setting)

Method	Validity	Cost per trial	Size	Power	Face validity	Status
1. F1, fixed 0.5	exact with respect to selection	1 fit	0.061; 0.051 in `docs/47`	0.400	weak	evaluated
F1A (AIC with the interaction)	none: post-selection	6 fits	0.074; 0.057 in `docs/47`	0.428 (inflated)	good	evaluated; not recommended
2. F1A0	approximate, first order	6 fits	0.067	0.396	good	illustrated
F1G, guarded F1A0	approximate; equals F1 unless 0.5 is rejected	6 fits	0.063	0.400	good	illustrated, post hoc
3. single split	exact with respect to selection	12 half-size fits	0.044	0.208	good	illustrated
3. cross-fit, average	approximate	12 half-size fits	0.061	0.372	good	illustrated
3. cross-fit, Cauchy	approximate	12 half-size fits	0.050	0.292	good	illustrated
4. profile interval	descriptive; coverage unknown	about 150 fits, 11 s	not a test	not a test	essential	illustrated, one trial
5. F1AB, bootstrap	by construction, up to the null model	M times F1A, minutes per trial	0.062	0.364	good	illustrated (warp-speed)

Method	Validity	Cost per trial	Size	Power	Face validity	Status
6. joint estimation	approximate; fails near boundaries	1 `nlme` fit and a sandwich	not run	not run	good	proposed
7. max-$\|t\|$, bootstrap	by construction, up to the null model	M times 5 fits	0.048	0.384	moderate: no single half-life	illustrated (warp-speed)
7. max-$\|t\|$, Bonferroni	exact, conservative	5 fits	0.024	0.220	moderate	illustrated
8. AIC-weight averaging	unknown; held size here	15 fits	0.053 (ℓ_1), 0.049 (ℓ_0)	0.388, 0.356	good	illustrated
8. Bayesian marginalization	unknown; calibration needed	MCMC, minutes	not run	not run	good	proposed
Permutation of B_i^c	invalid for a prognostic biomarker		not run	not run		not recommended
Residual permutation within path	approximate	M times the procedure	not run	not run	good	proposed

7.2 Provisional plan

Everything below is provisional. None of the data-chosen procedures has been evaluated at the scale of `docs/47`; the illustration has an MCSE of 0.007 for a pooled null rate and 0.031 for power, and the guarded rule F1G was defined after its results were seen.

1. **Primary analysis.** F1 with $t^a_{1/2}$ fixed in advance at 0.5 week, *guarded* by the data (F1G): fit the null model FM0 over $\mathcal{T}_a$; if its profile rejects 0.5 week ($2\{\max_h \ell_0(h) - \ell_0(0.5)\} > 3.84$), use the F1A0 choice $\widehat{h}_0$ instead; test with CR2. This meets the face-validity requirement as stated (the analysis never uses a half-life its data reject), behaved as F1 in the illustration (0.063 against 0.061 on the same trials, power 0.400), and departs from F1 in 1% to 15% of trials. It is preferred over plain F1A0, which retained about half of F1A's excess (0.067), and over cross-fitting, whose Cauchy version held size at a cost of about a quarter of the power (0.292 against 0.400).
2. **If the half-life must always be estimated** rather than defaulted, use AIC-weight averaging over the grid with null-model (ℓ_0) weights: it held size in every null cell (0.048 to 0.050 outside the prognostic cell) and lost 0.044 of power, within the Monte Carlo error. The ℓ_1-weighted version did as well here (0.053, power 0.388) but its weights see the interaction (Section 4.8).
3. **Calibration check.** Run F1AB and the bootstrap max-$|t|$ test by the warp-speed method in the stress-test extension, as the procedures valid by construction against which the cheap rules are judged. In the illustration the bootstrap brought F1A to F1's rate (0.062) but not to 0.05, because it cannot correct the null model's own misspecification.
4. **Reporting, in every case.** The null-model ML estimate of $t^a_{1/2}$, the profile likelihood over a fine grid with the deviance at 0, 0.5, 1 and 2 weeks, the profile-likelihood interval (described as such, with unknown coverage), the half-life actually used and why, and F1 over the grid as a sensitivity analysis. State that the analysis half-life is a working parameter and that a Hybrid trial distinguishes a short half-life from one of a week or more but not values within $[0, 0.5]$ week.
5. **Not to be used:** F1A, the max-$|t|$ statistic with a single-t reference, and naive permutation of the biomarker.

7.3 The stress-test extension needed to confirm it

- **Process.** The refined process (anchored carryover, randomized allocation, the driver's defaults), together with the pending anchored rerun of `docs/47`, so that F1 and the data-chosen rules are judged on the same trials.
- **Cells.** All 41 cells of `docs/46` (15 null at 1,000 replicates, 26 alternative at 500; 28,000 trials), plus 12 new cells crossing a wrong half-life with a prognostic biomarker: the constant and growing natural-course pathways and the placebo pathway, each at true $t_{1/2} = 0$ and 1 week, null (1,000) and $c_{bm} = 0.30$ (500); 9,000 more trials, 37,000 in all. The alternative cells with true $t_{1/2} = 0$ and 1 week are where a data-chosen half-life could recover power and must be included.
- **Arms.** F1, F1G, F1A0, AIC-weight averaging with ℓ_0 and ℓ_1 weights, cross-fit (average and Cauchy), F1AB and the max-$|t|$ test by warp-speed bootstrap, and the profile-interval coverage of the best-fitting grid value (estimated from a large noise-free fit per cell).
- **Cost.** The illustration's procedure took 7.1 worker-seconds per replicate with one bootstrap resample (Section 6.4), against about 2.1 for the driver's `--extra2` arm (`RERUN-anchored.md`, Section 3; quoted). At 7.1 seconds, 37,000 replicates need about 263,000 worker-seconds, 73 worker-hours: about 9 hours on 8 workers or 24 hours on the 3 cores of a default `simrun` job (derived; assumes the server's cores match this machine's, which is not measured).
- **Criteria.** Those of `docs/46` as amended: pooled null rate within the 2-MCSE band of 0.05 over the non-prognostic null cells, no cell above 0.075, prognostic cells judged separately, and power compared on paired trials.

8. Limitations

- **Stored setting.** Every stress-test number quoted here, and the illustration of Section 6, use configuration C at $\rho = 0.7$ with the published recursive carryover and fixed allocation. The refined process (anchored carryover, randomized allocation) has not been run at scale; Example 1 uses it for one trial only.
- **Illustration, not evaluation.** Section 6 has 250 to 500 replicates per cell in four cells (MCSE about 0.010 to 0.014 for a rate near 0.05), so differences of 0.01 between methods are not resolved, and the warp-speed rates carry additional Monte Carlo error from the estimated critical value.
- **Arguments, not proofs.** The validity arguments for F1A0 (Section 4.2) and the orthogonality result (Section 4.6) are first order. They do not cover a prognostic biomarker, informative dropout, or the boundary at $t^a_{1/2} = 0$, where the example trial's estimate lay.
- **Post hoc rules.** F1G and averaging with ℓ_0 weights were defined after the illustration's results were seen; the recommended primary rule is therefore the least tested of the candidates.
- **No power where it matters.** The only alternative cell run has the true half-life equal to the fixed value; the power gain or loss of a data-chosen half-life when 0.5 week is wrong is unknown.
- **One design, one sample size.** Hybrid at $N = 70$ only.
- **Not implemented:** joint estimation with a sandwich that includes $\partial D_{bc,it}/\partial\omega$ (Section 4.6), the multivariate normal calibration of the max-$|t|$ test (Section 4.7), the Bayesian analysis (Section 4.8), the residual permutation test (Section 4.9), and the profile interval's coverage (Section 4.4).

9. How to reproduce

From the repository root:

```
Rscript analysis/scripts/data-chosen-half-life/57-example-trial.R --cores 7 --boot 199
Rscript analysis/scripts/data-chosen-half-life/57-null-illustration.R --cores 7
Rscript analysis/scripts/data-chosen-half-life/57-posthoc-rules.R
Rscript analysis/scripts/data-chosen-half-life/57-stored-tables.R
```

Both source `57-common.R`, which evaluates the top-level definitions of `analysis/scripts/quick-sim/carryover-closed-fo` (everything before its pre-run checks, skipping the lines that create output directories), so the data-generating process and the fitting helpers are the driver's own.

10. Reproducibility

Every number labeled verified in Sections 1 to 7 was computed by the scripts in `analysis/scripts/data-chosen-half-life/`

- `57-common.R`: the procedures (grid fits, F1, F1A, F1A0, AIC-weight averaging, max-$|t|$, split and cross-fit, null-model bootstrap).
- `57-example-trial.R` and `57-example-figure.R`: Section 5, Tables 4 and 5 and Figure 1; outputs `57-example-*.csv` in `analysis/data/quick-sim/data-chosen-half-life/`.
- `57-null-illustration.R`: Section 6, Tables 6 to 8 and Figure 2; outputs `57-mc-*.csv`, `57-mc-replicates.rds` and the checkpointed chunks in `57-mc-chunks/`.
- `57-posthoc-rules.R`: the post hoc rows of Table 6 (F1G, averaging with ℓ_0 weights), the pooled null rates and the paired discordance counts; outputs `57-mc-posthoc-rules.csv`, `57-mc-pooled-null.csv` and `57-mc-discordance.csv`.

Tables 2 and 3 were computed by `57-stored-tables.R` from the stored stress-test replicates (`analysis/data/quick-sim/carryover-closed-form/strawman-stress/replicates-full.rds` and `replicates-full-extra2.rds`) by matching cell and replicate.

11. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013.
2. Pinheiro JC, Bates DM. *Mixed-Effects Models in S and S-PLUS*. New York: Springer; 2000. doi:10.1007/b98882.
3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*. 2002;28(2):169-181.
4. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics*. 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004.
5. Giacomini R, Politis DN, White H. A warp-speed method for conducting Monte Carlo experiments involving bootstrap estimators. *Econometric Theory*. 2013;29(3):567-589. doi:10.1017/S0266466612000655.
6. Davies RB. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*. 1977;64(2):247-254. doi:10.2307/2335690.
7. Davies RB. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*. 1987;74(1):33-43. doi:10.1093/biomet/74.1.33.
8. Cox DR. A note on data-splitting for the evaluation of significance levels. *Biometrika*. 1975;62(2):441-444. doi:10.1093/biomet/62.2.441.
9. Chernozhukov V, Chetverikov D, Demirer M, Duflo E, Hansen C, Newey W, Robins J. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*. 2018;21(1):C1-C68. doi:10.1111/ectj.12097.
10. Liu Y, Xie J. Cauchy combination test: a powerful test with analytic p-value calculation under arbitrary dependency structures. *Journal of the American Statistical Association*. 2020;115(529):393-402. doi:10.1080/01621459.2018.1554485.
11. Self SG, Liang KY. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*. 1987;82(398):605-610. doi:10.1080/01621459.1987.10478472.

12. Hansen BE. Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica*. 1996;64(2):413-430. doi:10.2307/2171789.
13. Buckland ST, Burnham KP, Augustin NH. Model selection: an integral part of inference. *Biometrics*. 1997;53(2):603-618. doi:10.2307/2533961.
14. Hoeting JA, Madigan D, Raftery AE, Volinsky CT. Bayesian model averaging: a tutorial. *Statistical Science*. 1999;14(4):382-417. doi:10.1214/ss/1009212519.
15. Freedman D, Lane D. A nonstochastic interpretation of reported significance levels. *Journal of Business and Economic Statistics*. 1983;1(4):292-298. doi:10.1080/07350015.1983.10509354.
16. Leeb H, Pötscher BM. Model selection and inference: facts and fiction. *Econometric Theory*. 2005;21(1):21-59. doi:10.1017/S0266466605050036.
17. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086.